Preeyansh Arora Majoka

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Education

University of British Columbia May 2026

B.A.Sc. Manufacturing Engineering | IMES Award 2021-2026 (Top 10% Faculty of Engineering) | Co-op Program

Relevant Courses: FEM, CFD, Vibration Mechanics, Advanced Manufacturing, Industrial Automation, Software System Design

Technical Skills

Mechanical: Creo, Windchill, SolidWorks, FEA, FEM using Python, Tolerance Stack-Up, DFM/DFA, GD&T, Ansys, DDM, SimScale

Manufacturing & Ops.: Siemens TIA Portal, CODESYS, PLC Programming (LD/ST), IFS/SAP ERP, CNC Machining, 3D Printing

Software & Hardware: Python (NumPy/Pandas/Tkinter), MATLAB, Excel (VBA, Power Query), Gurobi Optimizer, Git, PCB Design (KiCad), Hardware Debugging (Oscilloscope, DMM), Board Bring-up & Validation, LT Spice, Soldering

Experience

Mechatronics Engineer Intern - Product Development Team Jan 2025 – Sep 2025

ARTMS-Telix Pharmaceuticals Limited *Burnaby, BC*

- Designed **20+** tight-tolerance parts and **5+** subassemblies in **SolidWorks**, applying **DFM/DFA** to reduce part count by **20%** and improve assembly efficiency, and applying **GD&T** to ensure manufacturable, accurate assembly
- Developed a 4-layer voltage-controlled nickel electroplating PCB in **KiCad**, optimizing stack-up and grounding to mitigate EMI, regulate deposition rates, and reduce defects by **15%**, safeguarding **$80,000+** in monthly assets
- Led design reviews for USC Molecular Imaging Center, delivering a quick-mount hot-cell backplate using a **bayonet interface** that reduced installation time by **75%** and minimized radiation exposure to service personnel
- Engineered Gen3 Target Station prototype, reducing volumetric footprint by **70%** to meet Siemens cyclotron constraints
- Validated cooling-finger thermal performance in **SimScale**, maintaining temperature rise within $\pm$ **25 °C** during beam runs

Mechanical Engineer Intern - Vehicle Integration Team May 2024 – Dec 2024

Hexagon Purus Systems Canada *Kelowna, BC*

- Reduced dimensional variation by **25%** by combining **WC/RSS** tolerance stack-ups with Monte Carlo simulation using custom-built **Excel** models and **Creo EZ Tolerance** to identify dominant contributors (bolt-hole float, weld distortion)
- Redesigned structural brackets and used **FEA** to validate stress, displacement, and modal response under **3G loads** on **Creo Simulate** to prevent structural resonance, improving tooling access to the auxiliary module
- Consolidated HV phase cable and powertrain **CAN** harness inventory into **15 standard lengths** across **400+ truck configurations**, utilizing **Creo Cabling** and a **VBA-driven optimization model** to slash part variance
- Automated a fleet-wide Production Deviation Notice (PDN) tracker for **20+ trucks (50–80 PDNs each)** using **Excel** and **Power Query**, enhancing traceability and data access across Canadian and US teams
- Developed brake and lighting requirements for the Tern RC-8 truck by applying **FMVSS/CMVSS** motor vehicle standards

Design Team & Projects

President & Chassis Lead Jan 2025 – Present

OKMotorrad MotoStudent Team *Kelowna, BC*

- Founded UBCO's electric motorcycle team, recruiting **20+** members and establishing a multi-disciplinary structure
- Built a functional electric dirt bike prototype in **4 months** in under **$200** by repurposing scrap materials and fabricating custom brackets to drive an **1800W** BLDC motor, successfully validating feasibility to fast-track faculty accreditation
- Led frame topology research and prototype iterations in **SolidWorks Weldments**, simulating structural loads in **SolidWorks Simulate** and **Ansys Mechanical**, improving stiffness-to-weight ratio by $\sim$**15%**

Project Proposer & Manufacturing Lead Sep 2025 – Present

Target Capsule Feedback Capstone *Kelowna, BC*

- Defined ARTMS-Telix capstone requirements, led client meetings and used **MS Project** to align deliverables and timelines
- Designed, fabricated, and assembled a T&V bench using **SMC PneuDraw** and **SolidWorks** to simulate cyclotron pneumatic transfer, supporting synchronized **1–5 kHz** multi-sensor position tracking for radioisotope-carrying capsule
- Designed and prototyped a 3D-printed sensor-mounted coupler using **SolidWorks** for a **Ø1.5"** pneumatic hose, ensuring airtight splicing and enabling reliable capsule tracking over **50'-100'** of buried hose length
- Simulated force and torque profiles using **MATLAB** for an in-hose screw-driven robot to understand operating loads

Analog Circuit & PCB Designer Mar 2025 – Jun 2025

Klon Centaur Overdrive Guitar Pedal *Burnaby, BC*

- Simulated a Klon-style overdrive pedal in **LTspice**, achieving desired transparent gain at **23x lower cost** ($350 vs $8000)
- Designed schematic and PCB in **KiCad**, implementing **33 dB** multi-stage op-amp gain with germanium diode soft-clipping, buffered output on a **9 V** supply with $\sim$**18 V** internal rail for extended headroom, and a true-bypass circuit for clean tone
- Modeled a custom pedal enclosure and assembly in **SolidWorks**, 3D-printed a prototype, and assembled the final unit